

Production Pipeline – The Video Factory

A durable Inngest workflow. Each stage is a `step.run` that survives Vercel's function timeout, retries on failure, and writes a `production_steps` cost row when it touches a paid API. The cheap planning stage runs first so the cost gate has real numbers before any money is spent.

The 45-second structure

Every video targets 45s and follows the GAS narrative spine. The script, the scene cuts, AND the audio map are all built against these beats:

Section	~Time	Purpose
Hook	0–3s	Stop the scroll; name the problem
Insight	3–18s	The non-obvious truth
Example	18–32s	A concrete, relatable case
Resolution	32–40s	The shift / what to do
Closing	40–45s	A thought that creates curiosity

This bounds scene count ($\approx 4-6$) which bounds b-roll and render cost — the 45s target is a creative *and* cost decision.

Stages

```
brief.lock → script.generate →
cost.estimate → [GATE 1] →
audio.dynamic (TTS/uploaded + arc
music + per-scene ambient) →
aroll (HeyGen) || broll (Higgsfield) →
```

```
assemble.draft → [GATE 2] →  
assemble.final → [QA GATE] → deliver
```

1. **brief.lock** — operator picks the brain

(dropdown), topic, segment, and toggles:

popups (off by default; if on, choose style:

glass / solid / minimal), music,

ambient, resolution, aspect_ratio

(9:16 / 1:1 / 16:9 per

platform), higgsfield_tier, voice mode

(BYO voice_id vs uploaded audio). Lock

to structured brief JSON.

2. **script.generate** (Claude Opus, cheap) —

emits ONE JSON object:

- voiceover — the 45s script in the structure above
- scenes[] — per scene: visual direction, b-roll prompt (+ Soul id), avatar notes
- captions[] — {text, start_s, end_s}
- popups[] — {text, start_s, style} **only if** brief.popups is on
- audio_map — music arc + per-scene ambient + accent cues (see audio-

direction.md)

- cost metrics: char_count , duration_s , scene_count
- **Self-verifies** before emitting (hook in 3s, no banned words, human, every visual earns its place); pulls the client's winning patterns from the learning layer. Revises until it passes — never stops at draft one.

3. **cost.estimate** — metrics through

rate_card → estimate_cents . See cost-controls.md .

4. **GATE 1 (pre-flight)** — budget check (hard gate blocks if over) + operator approves the estimate. Nothing paid runs until this passes.

5. **audio.dynamic** — see audio-direction.md . Branch on voice mode:

- *BYO*: ElevenLabs TTS from voice_id → narration + word-level timestamps.
- *Uploaded*: use the file; run STT/alignment to recover word

timestamps (zero TTS spend).

In parallel, from `audio_map` (each only if enabled in the brief — SFX/ambient is off by

default): ElevenLabs Music v2 (arc-following track, client sonic

identity) + SFX (per-scene ambient

beds). Timestamps feed sync + caption alignment.

6. **aroll** — HeyGen Avatar V (standard API), **driven by the narration audio** so the client's voice carries the whole cut. Poll to completion.
7. **broll** — Higgsfield DoP per scene with the brain's Soul id for a consistent identity. **Fan out scenes in parallel.** Tier from `brief.higgsfield_tier`. The producer marks hero frame(s); run those through **Magnific Skin-Enhancer/Precision** (low creativity) for pore-level realism before animating — selective, to keep cost down.
8. **assemble.draft** — Shotstack JSON timeline at **low res / lite**: a-roll + b-roll +

captions + the full audio mix (music ducked under VO, ambient crossfades, pop-up accents)

- pop-up overlays (if any) + transitions.
Cheap proof for approval.

9. **GATE 2 (draft approval)** — operator reviews in the portal. Reject → revise only the relevant stage. Approve → continue. **You never pay for an HD render of a reject.**

10. **assemble.final** — same timeline, full resolution/quality → final MP4. (Audio is reused from the draft — only the visual render changes; see cost note.)

11. **QA GATE (integrity review)** — automated check on the finished cut, tiered severity:

- *Critical (block)*: banned word spoken/shown, wrong client identity/voice, off-brand logo/colors, duration far off 45s, missing captions.
- *Minor (flag, allow)*: pacing wobble, weak hook, caption timing nits.

Mirror your campaign-analyst integrity gate. Block on critical; surface minors to the operator.

12. **deliver** — store final in Blob

(`clients/{id}/...`), set `final_video_url` , mark `complete` . Portal shows preview + **Download**.

Iteration 2: publishing, Meta ad-loading, and logging to `published_assets` to feed the learning loop (`brains.md`). Not in Iteration 1 — this is not a Meta-integrated platform yet.

Pop-ups ("on screen pop-ups for effect, when asked")

Driven by `brief.popups` (off by default). When on, `script.generate` emits `popups[]` cues and matching `audio_map.accents` ; assemble adds **glass-box (glassmorphism)** animated overlays — frosted, brand-tinted from `clients.brand` (see `ux-flow.md`) — at the cued

timestamps, with the accent SFX firing on entrance. When off, the layer is skipped — no cost, no clutter.

Notifications (Gmail)

At GATE 1 and GATE 2, when an approval is needed, and at deliver when the final is ready, send a Gmail notification to the producer (and, on deliver, optionally a client hand-off email). Edge automations like this can run via Zapier off an Inngest event — but the pipeline itself stays in Inngest.

Cost logging

Every paid stage writes `production_steps` with `units` + `cost_cents` on completion, so `productions.actual_cents` reconciles live against `estimate_cents`.